

## 2025 AMC 10A Problem 6 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10A Problem 6. Great practice for AMC 10, AMC 12, AIME, and other math contests

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### 2025 AMC 10A Problem 6

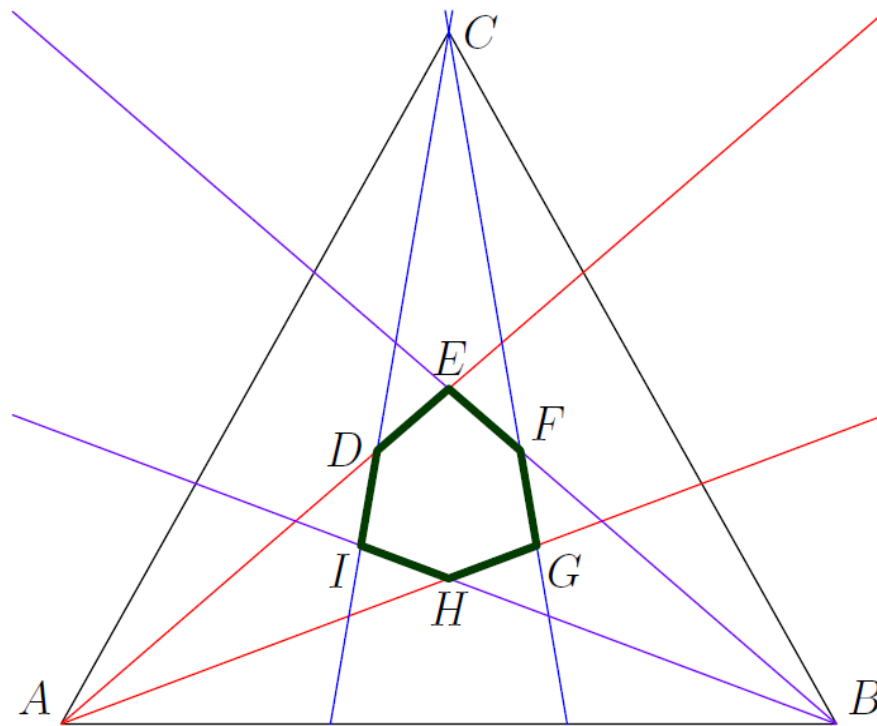
**Problem:**

In an equilateral triangle each interior angle is trisected by a pair of rays. The intersection of the interiors of the middle  $20^\circ$ -angle at each vertex is the interior of a convex hexagon. What is the degree measure of the smallest angle of this hexagon?

**Answer Choices:**

- A. 80
- B. 90
- C. 100
- D. 110
- E. 120

**Solution:**



Let  $ABC$  be an equilateral triangle with side  $AB$  horizontal. Each interior angle is  $60^\circ$  and is trisected into three  $20^\circ$  angles.

At  $A$  let the trisectors (starting from  $AB$  and going toward  $AC$ ) be the rays  $\ell_1, \ell_2$ ; at  $B$  the trisectors (from  $BA$  toward  $BC$ ) be  $m_1, m_2$ ; at  $C$  (from  $CA$  toward  $CB$ ) be  $n_1, n_2$ .

The middle  $20^\circ$ -angle at each vertex is the region between  $\ell_1, \ell_2$ , between  $m_1, m_2$ , and between  $n_1, n_2$  respectively.

Measure directions from the positive  $x$ -axis (along  $AB$ ):

- $AB$  has direction  $0^\circ$ , and  $AC$  has direction  $60^\circ$ . Hence  $\ell_1, \ell_2$  have directions

$$\angle(\ell_1) = 20^\circ, \quad \angle(\ell_2) = 40^\circ.$$

- At  $B$ , the sides  $BA, BC$  have directions  $180^\circ$  and  $120^\circ$ , so the trisectors are

$$\angle(m_2) = 160^\circ, \quad \angle(m_1) = 140^\circ.$$

► At  $C$ , the sides  $CA, CB$  have directions  $240^\circ$  and  $300^\circ$ , so

$$\angle(n_1) = 260^\circ, \quad \angle(n_2) = 280^\circ.$$

The hexagon is the intersection of the three middle  $20^\circ$  sectors; its sides lie on the six lines  $\ell_1, \ell_2, m_1, m_2, n_1, n_2$ . One checks from the picture that its vertices are the intersections

$$(\ell_1, m_2), (\ell_1, n_2), (\ell_2, m_1), (\ell_2, n_1), (m_1, n_1), (m_2, n_2).$$

The angle between two lines with directions  $\alpha, \beta$  is  $\min\{|\alpha - \beta|, 360^\circ - |\alpha - \beta|\}$ .

Thus for these pairs we get

$$\begin{aligned} (\ell_1, m_2) &: |20^\circ - 160^\circ| = 140^\circ, \\ (\ell_1, n_2) &: 360^\circ - |20^\circ - 280^\circ| = 100^\circ, \\ (\ell_2, m_1) &: |40^\circ - 140^\circ| = 100^\circ, \\ (\ell_2, n_1) &: 360^\circ - |40^\circ - 260^\circ| = 140^\circ, \\ (m_1, n_1) &: |140^\circ - 260^\circ| = 100^\circ, \\ (m_2, n_2) &: |160^\circ - 280^\circ| = 120^\circ \text{ (exterior), so interior} = 140^\circ. \end{aligned}$$

Hence the interior angles of the hexagon are three  $100^\circ$  angles and three  $140^\circ$  angles. Therefore the smallest angle of the hexagon is

(C) 100

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